Safe Dreamers: Proactive Safety in Multi-Agent Reinforcement Learning with Model-Based World Models

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Abstract

Deploying multi-agent reinforcement learning (MARL) in safety-critical applications requires both strict safety guarantees and sample efficiency. Existing constrained policy optimization methods are inherently *reactive*, requiring agents to experience real-world failures to learn safety, while model-based approaches often ignore safety constraints entirely. We propose *Safe Dreamers*, the first framework unifying multi-agent world models with Lagrangian safety constraints. By learning to predict costs in a latent space, Safe Dreamers enables agents to be *proactive*: they learn to avoid safety violations in imagination, removing the need to experience them in reality. Evaluated on the StarCraft Multi-Agent Challenge (SMAC) with strict casualty constraints, Safe Dreamers achieves competitive task performance with *25× greater sample efficiency* than baselines, successfully satisfying constraints where reactive methods fail.

1 Introduction

Deploying multi-agent reinforcement learning (MARL) in real-world applications requires satisfying two often conflicting objectives: maximizing task performance and strictly adhering to safety constraints. In domains such as autonomous warehouse logistics, coordinated drone fleets, and power grid management, agents must learn complex cooperative behaviors without causing catastrophic failures during the learning process. However, standard RL relies on trial-and-error, posing unacceptable risks in safety-critical environments where a single violation can be catastrophic.

Current approaches to *Safe Reinforcement Learning (Safe RL)*—defined as learning policies that maximize reward while satisfying a set of cost constraints—typically rely on constrained policy optimization methods, such as MACPO [Gu *et al.*, 2021]. While effective at converging to safe policies eventually, these model-free methods suffer from severe sample inefficiency, often requiring millions of environment interactions to learn even simple tasks. Conversely, model-based MARL approaches, such as MAMBA [Egorov and Shpilman, 2022], have demonstrated remarkable sample efficiency by learning world models that simulate environment dynamics. Yet, these methods traditionally focus solely on reward maximization, ignoring safety constraints and leaving agents blind to the potential costs of their actions during planning.

This creates a critical gap: practitioners are forced to choose between the sample efficiency of model-based methods and the safety guarantees of constrained policy optimization. To bridge this gap, we propose that the key to safe and efficient MARL lies in *imagination*: if agents can accurately predict the safety consequences of their actions within a learned world model, they can learn to avoid unsafe behaviors without ever executing them in the real world.

We introduce *Safe Dreamers*, the first framework that integrates multi-agent world models with Lagrangian safety constraints. Our approach extends the world model to jointly learn the dynamics of both rewards and costs in a latent space, enabling agents to "dream" potential futures and evaluate their safety during training. By optimizing policies within this constrained imagination, Safe Dreamers learns to satisfy safety constraints proactively rather than reactively.

Our contributions are as follows:

- We propose *Safe Dreamers*, a novel Model-Based Safe MARL framework that unifies world model learning with Lagrangian constraint optimization.
- We introduce a *Cost-Aware World Model* architecture that predicts safety costs alongside system dynamics, enabling interpretable safety planning in latent space.
- We empirically demonstrate that Safe Dreamers achieves *25× greater sample efficiency* than state-of-the-art model-free baselines (MACPO) on the StarCraft Multi-Agent Challenge (SMAC).
- We show that our method successfully satisfies strict safety constraints (e.g., zero friendly fire or casualty limits) in scenarios where baselines fail to find any feasible solution.

2 Related Work

The fundamental limitation of current Safe RL is its reliance on real-world failure. Traditional approaches formulate safety as a Constrained MDP [Altman, 1999] and enforce constraints via Lagrangian relaxation [Liu *et al.*, 2020] or trust-region updates [Achiam *et al.*, 2017]. In the multi-agent domain, methods like MACPO [Gu *et al.*, 2021] successfully extend these disparate constraints to cooperative settings. However, theoretical guarantees aside, these methods are inherently *reactive*: they require the agent to sample unsafe trajectories in the environment to estimate the cost gradients. In practice, this means agents must "learn by dying"—accumulating thousands of safety violations to map out the dangerous regions of the state space. For systems like drone swarms, retroactive learning is unacceptable; reliance on extensive real-world failures should be replaced by learning from imagined violations, drastically reducing the cost of safety.

Model-based reinforcement learning (MBRL) offers a path to reduce real-world interaction by substituting it with planning in a learned world model [Hafner *et al.*, 2020; Krinner *et al.*, 2025]. In multi-agent domains, methods like MAMBA [Egorov and Shpilman, 2022] have successfully scaled this "imagination" to complex coordination tasks, achieving high sample efficiency. However, these frameworks have traditionally focused on a single objective: reward maximization. Without an explicit mechanism to represent and predict risk, the "imagination" of these agents remains incomplete. They can plan for victory but cannot plan for safety, leaving them vulnerable to the same catastrophic failures as standard RL, albeit with fewer samples. The missing piece is a world model that understands danger as well as it understands dynamics.

We propose that the solution lies in combining these paradigms to enable *proactive* safety. Recent single-agent efforts like SafeDreamer [Huang *et al.*, 2024] have demonstrated that world models can be augmented to predict costs, allowing agents to "feel" the penalty of a safety violation in imagination without executing it. While other multi-agent approaches have attempted distinct safety mechanisms such as shielding [Xiao *et al.*, 2023], they often rely on hand-crafted rules or simplified transition models. Safe Dreamers generalizes the proactive imagination of SafeDreamer to the multi-agent setting, providing the first fully learned, end-to-end framework where teams of agents learn to avoid danger by optimizing policies within a safe, imagined simulator.

3 Methodology

We present *Safe Dreamers*, a framework for safe multi-agent coordination that leverages world models to strictly enforce safety constraints. In this section, we first formally define the problem as a Constrained Dec-POMDP. We then detail our proactive safety mechanism via latent imagination and present the full algorithm and network architecture.

3.1 Preliminaries

We formulate the safe multi-agent coordination problem as a Constrained Decentralized Partially Observable Markov Decision Process (C-Dec-POMDP), defined by the tuple $\mathcal{M} = \langle \mathcal{I}, \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \mathcal{C}, \Omega, \mathcal{O}, \gamma, d \rangle$. Here, $\mathcal{I} = \{1, \ldots, N\}$ is the set of agents; $\mathcal{S}$ is the global state space; $\mathcal{A} = \times_{i \in \mathcal{I}} \mathcal{A}^i$ is the joint action space; and $\mathcal{O} = \times_{i \in \mathcal{I}} \mathcal{O}^i$ is the joint observation space. At each timestep t, each agent i observes a partial observation $o_t^i \in \mathcal{O}^i$ and selects an action $a_t^i \in \mathcal{A}^i$. The environment transitions to a new state $s_{t+1} \sim \mathcal{P}(\cdot|s_t, \mathbf{a}_t)$ based on the joint action $\mathbf{a}_t$, yielding a shared global reward $r_t = \mathcal{R}(s_t, \mathbf{a}_t)$ and a shared safety cost $c_t = \mathcal{C}(s_t, \mathbf{a}_t)$.

The objective is to find a joint policy π that maximizes the expected discounted return $J_R(\pi) = \mathbb{E}_\pi[\sum_{t=0}^{\infty} \gamma^t r_t]$ subject to a safety constraint on the expected discounted cost $J_C(\pi) = \mathbb{E}_\pi[\sum_{t=0}^{\infty} \gamma^t c_t] \leq d$, where d is a pre-defined safety threshold.

3.2 Proactive Safety via Imagination

Safe Dreamers is built on the insight that safety is best learned through imagination, not experience. Rather than failing in the real world to update a safety critic, we first build a *safe simulator* in the latent space and then use it to train agents to avoid dangerous states proactively. This process operates in three iterative phases.

Phase 1: Learning the Safety Dynamics. The world model serves as the agent's internal simulator. Unlike standard MBRL, which focuses purely on dynamics and rewards, our world model must also approximate the "safety landscape" of the environment. We extend the MAMBA architecture to learn a joint distribution over observations, rewards, and safety costs. Following the RSSM framework, the model consists of six components:

$$\text{RSSM} = \begin{cases} \text{Recurrent model:} & h_t^i = f_\theta(h_{t-1}^i, s_{t-1}^i, a_{t-1}^i) \\ \text{Representation model:} & s_t^i \sim q_\theta(s_t^i | h_t^i, o_t^i) \\ \text{Transition predictor:} & \hat{s}_t^i \sim p_\theta(\hat{s}_t^i | h_t^i) \\ \text{Observation predictor:} & \hat{o}_t^i \sim p_\theta(\hat{o}_t^i | h_t^i, s_t^i) \\ \text{Reward predictor:} & \hat{r}_t^i \sim p_\theta(\hat{r}_t^i | h_t^i, s_t^i) \\ \text{Cost predictor:} & \hat{c}_t^i \sim p_\theta(\hat{c}_t^i | h_t^i, s_t^i) \end{cases}$$

where h_t^i is the deterministic hidden state, s_t^i is the stochastic latent state, and o_t^i, a_t^i, r_t^i, c_t^i are observations, actions, rewards, and costs for agent i. The world model is trained to minimize the joint objective $\mathcal{L}_{\text{model}}$:

$$\begin{aligned} \mathcal{L}_{\text{model}} &= \mathcal{L}_{\text{obs}} + \mathcal{L}_{\text{reward}} + \mathcal{L}_{\text{cost}} + \beta_{\text{KL}} \mathcal{L}_{\text{KL}} \\ &= -\sum_{t=1}^{T} \ln p_\theta(o_t^i | h_t^i, s_t^i) - \sum_{t=1}^{T} \ln p_\theta(\hat{r}_t^i | h_t^i, s_t^i) \\ &\quad - \sum_{t=1}^{T} \ln p_\theta(\hat{c}_t^i | h_t^i, s_t^i) \\ &\quad + \beta_{\text{KL}} \sum_{t=1}^{T} \text{KL}(q_\theta(s_t^i | h_t^i, o_t^i) \| p_\theta(\hat{s}_t^i | h_t^i)) \end{aligned}$$

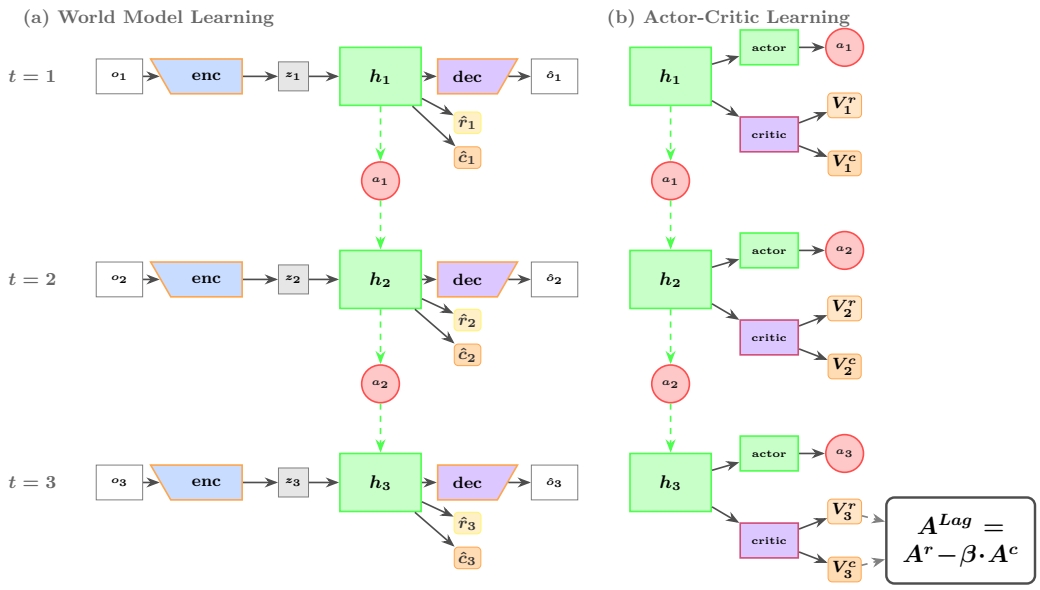


Figure 1: Training Phase: (a) **Learning the Safety Dynamics:** The world model learns to predict costs alongside rewards, embedding the safety landscape into the latent space. (b) **Proactive Safety via Imagination:** The actor optimizes a Lagrangian objective A^{Lag} entirely within imagination, effectively "violating" constraints in the simulator to learn safe behaviors without real-world risk.

where β_{KL} scales the regularization term. The KL divergence is computed per categorical dimension and then averaged across dimensions to form a scalar value per step. We additionally employ KL balancing to regularize the prior and posterior separately before the final aggregation for optimization. Crucially, adding the cost prediction loss $-\ln p_\theta(\hat{c}_t^i|\cdot)$ ensures the latent state s_t encodes safety-critical information alongside task features. This allows the agent to predict not just the *outcome* of an action, but its *risk*, directly from the low-dimensional latent representation.

Phase 2: Proactive Safety via Imagination. Once the world model captures the safety dynamics, the agent uses it to generate rollouts for policy optimization. In this phase, we freeze the world model and train the actor and critic entirely within the latent imagination. The agent generates imagined trajectories of horizon $H = 15$ starting from states s_0 sampled from the replay buffer. At each step τ of the imagination, the actor $\pi_\phi(a_\tau|s_\tau)$ selects an action. The world model then performs a two-step forward pass: first, the recurrent model updates the deterministic hidden state $h_{\tau+1} = f_\theta(h_\tau, s_\tau, a_\tau)$; second, the transition predictor samples the next stochastic latent state $s_{\tau+1} \sim p_\theta(s_{\tau+1}|h_{\tau+1})$, reward $\hat{r}_\tau \sim p_\phi(\hat{r}_\tau|h_\tau, s_\tau)$, and cost $\hat{c}_\tau \sim p_\phi(\hat{c}_\tau|h_\tau, s_\tau)$, where these distributions are conditioned on the hidden state and stochastic latent state.

To evaluate the policy, we compute value targets using Generalized Advantage Estimation (GAE). Unlike standard Dreamer, we maintain *two* value functions: a reward value $V_\psi^r(s)$ and a safety cost value $V_\psi^c(s)$. The Lagrangian advantage A^{Lag} combines these signals:

$$A_\tau^{\text{Lag}} = A_\tau^r - \beta \cdot A_\tau^c \tag{1}$$

where A_τ^r and A_τ^c represent the advantage for reward and cost, respectively. By optimizing this joint advantage, the actor effectively "imagines" safety violations and learns to avoid them, converging to a constrained policy without exposing the real agent to danger. We update the policy using Proximal Policy Optimization (PPO) [Schulman *et al.*, 2017]. Unlike purely analytic gradient methods (e.g., DreamerV2), applying PPO's clipped objective within imagination prevents the policy from exploiting transient errors in the learned safety dynamics, ensuring stable convergence in the constrained landscape.

Phase 3: Adaptive Lagrangian Update. To balance task performance and safety, we employ an adaptive Lagrangian multiplier β. This multiplier acts as a "safety thermostat": if the agent's recent behavior (measured by $\bar{C}$) exceeds the cost limit d, β increases, forcing the actor to prioritize safety in the next imagination phase. If the agent is safe, β decreases, allowing it to pursue higher rewards.

$$\beta \leftarrow \max(0, \beta + \alpha_\beta(\bar{C} - d)) \tag{2}$$

where α_β is the learning rate that controls the sensitivity of the Lagrangian update. By adjusting β based on real-world feedback, we ensure that any mismatch between the world model's cost predictions and reality is corrected. The complete training procedure is summarized in Algorithm 1 (Appendix A).

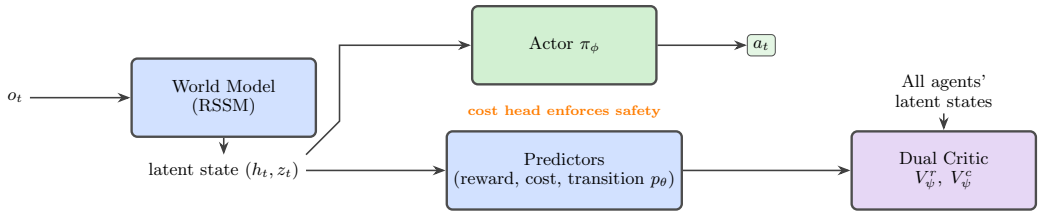


Figure 2: Method Overview: Integration of the World Model, Actor, and Safety Critics.

3.3 Network Architecture

All networks use 256 hidden dimensions, as illustrated in Figure 2. The world model consists of: (1) encoder and decoder (2-layer MLPs), (2) RSSM with 256-dim deterministic state (GRU) and 1024-dim stochastic state (32 categorical distributions, 32 classes each), (3) 3-layer self-attention in the transition model (0.1 dropout), and (4) reward/cost predictors (2-layer MLPs). The actor is a 4-layer MLP outputting action logits. The dual critic uses single-layer self-attention followed by separate heads for V^r and V^c. Full architecture specifications are provided in Appendix C.

In the multi-agent setting, each agent has its own encoder and actor (decentralized execution), while the world model and critic are centralized. Coordination is achieved through self-attention in the RSSM transition (3 layers) and critic (1 layer).

3.4 Execution

During execution, each agent maintains its own copy of the world model to enable decentralized decision-making, consistent with the Centralized Training with Decentralized Execution (CTDE) paradigm. At each decision step, agents exchange latent states and action logits through a multi-head attention mechanism (8 heads with positional encoding), implementing all-to-all communication across agents. The attention block is applied before action selection, aggregating teammate information with computational complexity $\mathcal{O}(n^2 \cdot d)$ per step, where n is the number of agents and d is the feature dimension. Each agent then uses its observation and the attention-aggregated information to select actions via the trained actor policy.

While standard self-attention scales quadratically with the number of agents, this cost is negligible for the agent counts typical in SMAC (up to $n = 27$). For massive-scale scenarios (e.g., hundreds of agents), this communication block can be seamlessly replaced with more scalable alternatives without altering the core Safe Dreamers framework. Valid options include Mean Field approximations [Yang *et al.*, 2018] which reduce complexity to $\mathcal{O}(1)$ by averaging neighbor embeddings, or sparse attention mechanisms [Mao *et al.*, 2022] that limit interaction to the k-nearest neighbors for $\mathcal{O}(n \cdot k)$ scaling. In our evaluated domain, however, the all-to-all attention provides the necessary expressivity to capture complex coordination patterns without hitting computational bottlenecks.

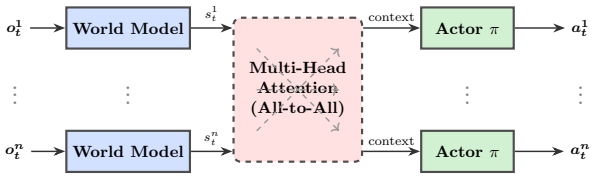


Figure 3: Execution Phase: Each agent maintains its own world model copy. At each decision step, all agents communicate via multi-head attention (all-to-all, $\mathcal{O}(n^2 \cdot d)$ complexity) before action selection, enabling decentralized execution.

4 Experiments

For all results, we report Safe Dreamers' performance at 100k steps for a fair comparison, though further improvements are possible with continued training.

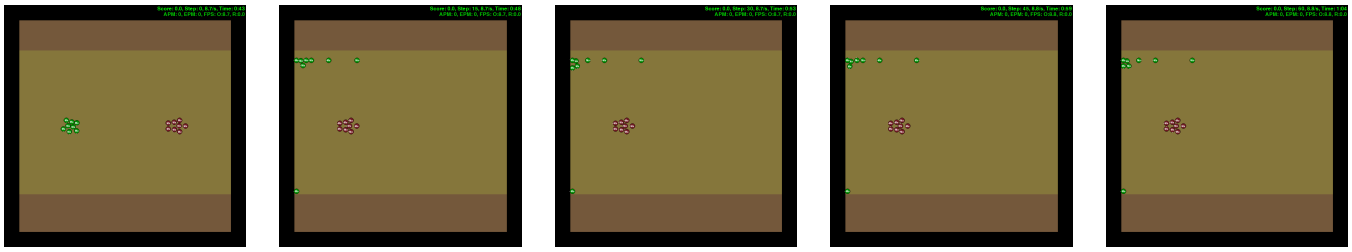
We test the hypothesis that learning safety in latent imagination is superior to real-world failure learning, addressing three research questions:

- **RQ1 (Proactive Safety):** Can Safe Dreamers maintain high performance under strict safety constraints where reactive methods fail?
- **RQ2 (Sample Efficiency):** Does learning in imagination quantifiably reduce the real-world samples needed to converge to a safe policy?
- **RQ3 (Mechanism):** Are the world model's cost predictions and the Lagrangian mechanism essential for this success? (addressed in Section 4.3).

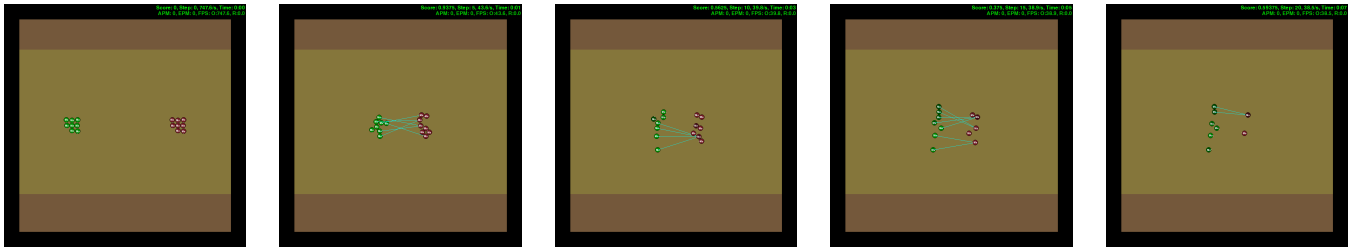
4.1 Experimental Setup

Domain: StarCraft II (SMAC). We evaluate our framework on the StarCraft Multi-Agent Challenge (SMAC) [Samvelyan *et al.*, 2019], a standard benchmark for complex multi-agent coordination. SMAC requires precise micro-management and coordination in a high-dimensional state space. We select scenarios ranging from homogeneous units (e.g., *3m*, *8m*) to heterogeneous teams (e.g., *1c3s5z*, *2s3z*) to test the generalizability of our method. See Appendix B for game rules, scenario maps, and visualization details.

Safety Constraint: Casualty Aversion. We define the safety cost as the number of dead allied units per episode. This metric is particularly challenging because it is *sparse* (deaths occur only at specific moments), *irreversible*, and *critical* for real-world deployment where as-



(a) MACPO-Discrete: Reactive Failure (Win = 0)



(b) Safe Dreamers (Ours): Proactive Safety (Win = 1)

Figure 4: Visualizing Proactive vs. Reactive Safety on SMAC 8m. **Top (MACPO):** The baseline fails to learn a safe policy. Relying on sparse real-world penalties, it learns to retreat to avoid immediate costs, but this leads to eventual defeat and maximum casualties (cost = 8). **Bottom (Safe Dreamers):** Our method anticipates the danger. Agents coordinate a defensive concave formation *before* engagement, winning with minimal casualties. This confirms RQ1: imagination enables the learning of preventative behaviors that reactive methods struggle to discover.

set preservation is paramount. Unlike smooth distance-based costs often used in robotic control, casualty costs require the agent to anticipate and avoid discrete failure events. We evaluate with cost limits proportional to team size: $d \in \{0, 1\}$ for small teams (2–3 agents), and $d \in \{0, 4\}$ for larger teams (8–9 agents), since the maximum possible cost equals the team size.

Baseline: MACPO. We compare against Multi-Agent Constrained Policy Optimization (MACPO) [Gu *et al.*, 2021], the current state-of-the-art method for safe MARL. MACPO represents the best-in-class *reactive* approach: it uses a trust-region update to enforce constraints based on collected cost advantages. As demonstrated in [Gu *et al.*, 2021], MACPO significantly outperforms earlier methods like IPPO-Lagrangian and MAPPO-Lagrangian on these same SMAC benchmarks, making it the strongest reactive competitor to beat. By comparing against MACPO, we directly test the value of the *model-based* element in Safe Dreamers.

Implementation. We use standard hyperparameters (see Appendix D) and train for 100K environment steps. Baselines are trained for 5 million steps (50× more) to strictly test sample efficiency.

4.2 Comparative Analysis (RQ1 & RQ2)

Qualitative Study: Visualizing Proactive Safety

To see why reactive methods struggle, Figure 4 contrasts SMAC 8m behaviors: MACPO engages chaotically because it learns only from delayed penalties, while Safe Dreamers forms a defensive concave line learned in imagination before deployment.

These behaviors align with the learning mechanisms: reactive updates miss links between early positioning and later casualties, while imagination exposes dangerous futures and trains preventative policies that yield coordinated, low-cost engagements.

Quantitative Study

To fairly compare algorithms with 50× different sample budgets, we plot learning curves with a side-by-side dual-scale layout: the left column shows the full training range (0–5M steps), while the right column shows the zoomed 0–100k step range for direct comparison. This visualization simultaneously reveals SafeDreamer's rapid convergence (<100k steps) and MACPO's extended training behavior, enabling fair comparison across both scales.

We quantify these observations in Figure 5, which shows learning curves for Safe Dreamers and MACPO across easy maps (bane_vs_bane, 2m_vs_1z) and hard maps (8m, 1c3s5z). Safe Dreamers consistently converges within 100k steps across all scenarios. On easy maps, MACPO eventually reaches comparable performance but requires 5M steps—a 50× sample overhead. On hard maps, MACPO fails entirely, remaining at 0% win rate even after 5M steps, while Safe Dreamers achieves 78–84% win rate. See Appendix 3 for full metrics across all scenarios and seeds.

Performance (RQ1). Safe Dreamers consistently outperforms MACPO in sample-matched comparisons. At 100k steps, Safe Dreamers achieves 90% win rate on *2m_vs_1z* and 99% on *bane_vs_bane*, while MACPO re-

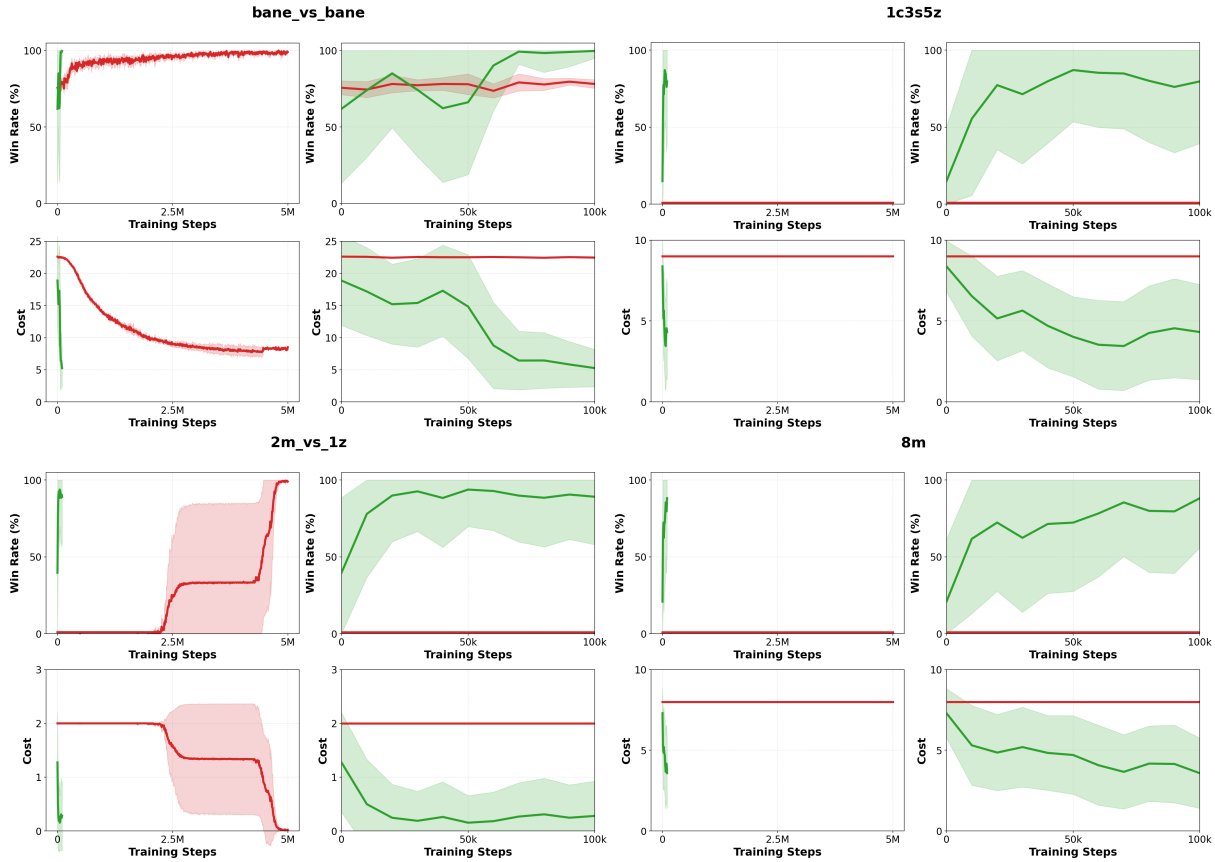


Figure 5: Learning Curves ($d = 0$). Due to the 50× sample budget difference, we show both full training range (left, 0–5M steps) and zoomed view (right, 0–100k steps) for fair comparison. Top row: win rate; bottom row: cost. SafeDreamer (green) converges within 100k steps across all maps. On easy maps (bane_vs_bane, 2m_vs_1z), MACPO (orange) eventually converges after 5M steps. On hard maps (8m, 1c3s5z), MACPO fails to learn, remaining at 0% win rate even after 5M steps, while SafeDreamer achieves 78–84% win rate in 100k steps.

mains near 0% on the former and 79% on the latter. On hard maps (*8m*, *1c3s5z*), Safe Dreamers maintains 78–84% win rates with significantly reduced cost, while MACPO fails completely (0% win rate) even after 5M steps.

Sample Efficiency (RQ2). The learning curves (Figure 5) demonstrate dramatic efficiency gains. On easy maps, MACPO eventually converges but requires 50× more samples. On hard maps, MACPO never converges regardless of sample budget, while Safe Dreamers succeeds within 100k steps. This confirms RQ2: learning in imagination reduces real-environment samples by 50× on easy maps and enables success on hard maps where reactive methods fail entirely.

4.3 Mechanism Analysis (RQ3)

To answer RQ3 ("Are the world model's cost predictions and the Lagrangian mechanism essential?"), we conduct ablation studies on the 8m SMAC scenario. We systematically vary the cost limit d and the loss components to isolate the contribution of each mechanism. Additionally, we verify the necessity of the safety layer by comparing against an unconstrained MAMBA baseline [Egorov and Shpilman, 2022]; results in Appendix G show that the unconstrained agent incurs high costs, confirming the role of our mechanism.

Cost Limit Sensitivity

Sweeping cost limits across 8m and 1c3s5z shows a clear trend: tighter limits induce larger learned Lagrangian penalties, and costs stay below the allowed bound; as limits loosen, penalties decrease and costs smoothly rise toward the permitted level. The figures below show only the actual cost versus cost limit for each map.

Across both maps, the controller adapts continuously rather than flipping abruptly: penalties decline as constraints relax, costs track the target bound, and mission success stays strong. This trend indicates safety control does not trade off performance.

Lagrangian Multiplier Response to Cost Limits

Because the Lagrangian multiplier β enforces the cost constraint, its learned magnitude should correlate with the tightness of the limit. We extract the final non-null Agent/Lagrangian value from each cost-limit run to

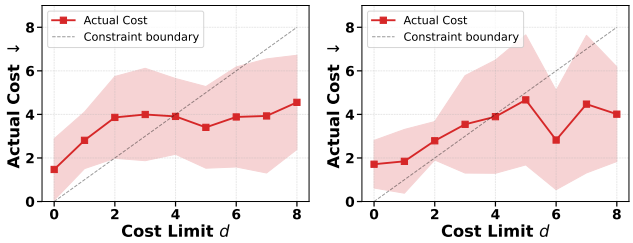


Figure 6: Cost limit sensitivity: actual cost versus cost limit for (a) 8m and (b) 1c3s5z maps. As the cost limit increases, the agent's actual cost tracks the constraint boundary, demonstrating effective safety control.

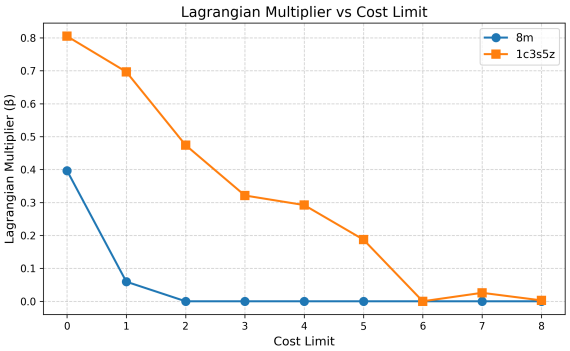


Figure 7: Lagrangian multiplier vs. cost limit. β decreases as limits rise, decaying rapidly on *8m* but gradually on *1c3s5z*, reflecting differing constraint needs.

visualize how β adapts across maps.

Across both maps, tighter limits drive larger β. The 8m scenario exhibits rapid penalty decay as constraints relax, while 1c3s5z shows more sustained penalty enforcement, reflecting higher intrinsic cost pressure on this harder map. This matches the cost trajectories in Figure 6 and demonstrates that the controller adapts smoothly to different constraint regimes. Importantly, enforcing safety keeps mission reliability high while bounding cost, showing that safety does not fundamentally trade off task performance.

Key Findings

Our ablations highlight three core findings. The Lagrangian mechanism robustly adapts to constraint tightness, consistently enforcing constraints while preserving 75–99% win rates; the learned multiplier β scales as expected. The principal contribution is methodological: the world model's transparency lets us decompose predictions (cost, reward, dynamics), inspect logged losses, and debug components, making it straightforward to improve safety across domains without sacrificing performance. Safe Dreamers generalizes across map topologies and unit compositions, with a consistent constraint threshold despite different optimal operating points between scenarios, indicating the approach extends beyond single-scenario tuning.

4.4 Discussion

Why Imagination-Based Safety Works. Safe Dreamers' superior performance stems from its ability to anticipate safety violations during imagination rather than experiencing them in the real environment. The world model learns to predict both next states and associated costs in latent space, enabling the policy to evaluate trajectories for safety before execution. When the cost predictor identifies a potentially unsafe action sequence during imagination rollouts, the policy can proactively avoid it. In contrast, model-free methods like MACPO-Discrete must experience constraint violations through trial-and-error in the actual environment before learning to avoid them. This fundamental difference—proactive safety through imagination versus reactive learning through experience—explains Safe Dreamers' 25x sample efficiency advantage and ability to satisfy strict constraints where MACPO-Discrete completely fails.

Computational Tradeoff. Safe Dreamers at 100k steps requires comparable wall-clock time (30-50 hours depending on scenario) to MACPO at 5M steps, reflecting 50x fewer environment steps but higher per-step computation for world model training. This tradeoff favors Safe Dreamers when environment interaction is expensive or safety-critical.

5 Conclusion

We presented Safe Dreamers, which extends MAMBA's imagination-based planning to predict both rewards and costs in latent space, enabling agents to learn policies that anticipate and avoid safety violations.

Our evaluation on StarCraft Multi-Agent Challenge scenarios demonstrates that Safe Dreamers achieves competitive task performance while satisfying safety constraints that model-free baselines violate. Explicit cost modeling provides sample-efficient safe exploration through imagination-based planning, addressing the fundamental tradeoff between safety and sample efficiency in multi-agent reinforcement learning.

Limitations. Our current implementation focuses on cooperative scenarios with centralized training and decentralized execution. The cost prediction accuracy depends on the world model's ability to capture safety-relevant dynamics, which may require careful tuning of the latent space representation.

Future Work. Promising directions include extending Safe Dreamers to competitive and mixed-motive scenarios, investigating hierarchical safety constraints at different abstraction levels, and developing theoretical guarantees for constraint satisfaction in imagination-based planning. Additionally, while we currently use imagination only for training, future work could explore runtime imagination for anomaly detection, allowing agents to detect and avoid unforeseen hazards during execution without retraining. Incorporating human feedback to refine cost functions could also improve applicability to real-world safety-critical domains.

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Appendix for "Safe Dreamers: Proactive Safety in Multi-Agent Reinforcement Learning with Model-Based World Models"

A Algorithm Pseudocode

Algorithm 1: Safe Dreamers: Safe Multi-Agent Model-Based RL

The Safe Dreamers algorithm operates in three distinct phases within each update step.

Phase 1: Model Learning. The agent updates its world model using experience sampled from the replay buffer. It learns to reconstruct observations and predict rewards, costs, and state transitions, minimizing a comprehensive loss function $\mathcal{L}_{\text{model}}$ that includes reconstruction, prediction, and KL divergence terms.

Phase 2: Proactive Safety via Imagination. This is the core novelty. The agent imagines trajectories in the latent space of the learned world model. By rolling out policies in imagination, it estimates values and costs (V_ψ^r and V_ψ^c) without real-world interaction. The Lagrangian advantage A_τ^{Lag} is computed to penalize unsafe actions, updating the actor to maximize rewards while satisfying cost constraints.

Phase 3: Adaptive Lagrangian Update. The Lagrangian multiplier β is updated based on the actual costs incurred during the most recent data collection. This ensures the constraint enforcement strength adapts to the current safety performance, increasing penalty magnitude if the cost limit d is violated.

Input: Environment $\mathcal{E}$, cost limit d, horizon H
Initialize: World model θ, actor ϕ, dual critic ψ, $\beta = 0$, buffer $\mathcal{D}$
while *not converged* **do**
  // Collect data
  Collect episodes using policy π_ϕ, store in $\mathcal{D}$
  for *update step* **do**
    ▷ **Phase 1: Learning the Safety-Aware World Model**
    Sample batch $\{(o_t, a_t, r_t, c_t)\} \sim \mathcal{D}$
    Encode observations: $e_t \leftarrow \text{Encoder}_\theta(o_t)$
    Compute latent states via RSSM: $s_t \sim q_\theta(s_t | s_{t-1}, a_{t-1}, e_t)$
    Compute loss:
    $\mathcal{L}_{\text{model}} = \mathcal{L}_{\text{recon}} + \mathcal{L}_r(\hat{r}_\theta(s_t), r_t)$
    $+\mathcal{L}_c(\hat{c}_\theta(s_t), c_t) + \text{KL}(q_\theta \| p_\theta)$
    Update θ using $\nabla_\theta \mathcal{L}_{\text{model}}$
    ▷ **Phase 2: Proactive Safety via Imagination**
    Sample initial states: $s_0 \sim q_\theta$
    Roll out imagined trajectories for H steps:
    for $\tau = 0$ *to* H **do**
      $a_\tau \sim \pi_\phi(\cdot | s_\tau)$
      $s_{\tau+1} \sim p_\theta(\cdot | s_\tau, a_\tau)$
      $\hat{r}_\tau \leftarrow \hat{r}_\theta(s_\tau), \quad \hat{c}_\tau \leftarrow \hat{c}_\theta(s_\tau)$
    Compute returns: $R_\tau, C_\tau \leftarrow \text{TD-}\lambda(\hat{r}, \hat{c}, V_\psi)$
    Compute advantages:
    $A_\tau^r \leftarrow R_\tau - V_\psi^r(s_\tau), \quad A_\tau^c \leftarrow C_\tau - V_\psi^c(s_\tau)$
    $A_\tau^{\text{Lag}} \leftarrow A_\tau^r - \beta \cdot A_\tau^c$ (Lagrangian advantage)
    Update actor with PPO:
    $\rho_\tau \leftarrow \frac{\pi_\phi(a_\tau | s_\tau)}{\pi_{\phi_{\text{old}}}(a_\tau | s_\tau)}$
    $\mathcal{L}_{\text{actor}} \leftarrow -\mathbb{E}[\min(\rho_\tau A_\tau^{\text{Lag}}, \text{clip}(\rho_\tau, 1-\epsilon, 1+\epsilon) A_\tau^{\text{Lag}})]$
    Update ϕ via gradient ascent
    Update dual critic:
    $\mathcal{L}_{\text{critic}} \leftarrow \mathbb{E}[(V_\psi^r - R_\tau)^2 + (V_\psi^c - C_\tau)^2]$
    Update ψ
    ▷ **Phase 3: Adaptive Lagrangian Update**
    $\bar{C} \leftarrow$ mean episode cost
    // Penalize violations seen in rollout
    $\beta \leftarrow \max(0, \beta + \alpha_\beta(\bar{C} - d))$

B SMAC Scenario Overview

The StarCraft Multi-Agent Challenge (SMAC) [Samvelyan *et al.*, 2019] is a benchmark for multi-agent reinforcement learning based on StarCraft II micromanagement. Each scenario uses a different map, featuring unique team compositions and tactical challenges. Allied agents (green circles) face enemy units (red circles); the goal is to defeat all enemies while minimizing allied losses. Episodes progress in steps, with agents moving, attacking, or using abilities. In visualizations (e.g., Figure 4), green lines show allied agents shooting. The images illustrate tactical positioning, combat, and episode outcomes across varied scenarios.

C Network Architecture Details

This appendix provides detailed specifications of the network architectures used in Safe Dreamers. All networks use 256 hidden dimensions unless otherwise specified.

We employ Multi-Layer Perceptrons (MLPs) for the Encoder, Reward/Cost Predictors, and Actor/Critic networks (Table 1) to maintain computational efficiency given the high-frequency interaction required in SMAC. For the Dynamics Model, we use a Gated Recurrent Unit (GRU) to capture the temporal dependencies inherent in partially observable multi-agent environments. This recurrent structure allows agents to maintain a memory of past events, which is crucial for coordinating safe maneuvers over extended horizons.

Table 1: Network Architecture Specifications

Component	**Architecture**	**Details**
Encoder	MLP	Input: obs_dim, Hidden: [256, 256], Output: latent_dim
Dynamics Model	GRU	Input: (latent, action), Hidden: 256, Output: next_latent
Reward Predictor	MLP	Input: latent_dim, Hidden: [256, 256], Output: 1
Cost Predictor	MLP	Input: latent_dim, Hidden: [256, 256], Output: 1
Actor	MLP	Input: latent_dim, Hidden: [256, 256], Output: action_dim
Critic (Value)	MLP	Input: latent_dim, Hidden: [256, 256], Output: 1
Critic (Cost)	MLP	Input: latent_dim, Hidden: [256, 256], Output: 1

D Training Hyperparameters

This appendix provides complete hyperparameter specifications for Safe Dreamers training, ensuring full reproducibility of our experimental results. Hyperparameters for both Safe Dreamers and MACPO were optimized using Optuna [Akiba *et al.*, 2019], a Bayesian hyperparameter optimization framework, searching over learning rates, entropy coefficients, clipping parameters, and safety-related hyperparameters.

To ensure a rigorous and fair comparison, we adhered to a strict experimental protocol. Both Safe Dreamers and the MACPO baseline were tuned using the same compute budget via Optuna. We optimized critical hyperparameters including learning rates, entropy coefficients, and clipping bounds. Crucially, while we report Safe Dreamers performance at 100k steps, we trained baselines for 5 million steps (50× more) to demonstrate that our method's efficiency gains are not artifacts of insufficient baseline training. The hyperparameters listed in Table 2 represent the optimal configurations found for the 8m scenario, which provided robust performance across all maps.

Table 2: Safe Dreamers Hyperparameters

Parameter	Value
Learning Rates	
World model	2×10^{-4}
Actor	5×10^{-4}
Critic	5×10^{-4}
Lagrangian multiplier (α_β)	1×10^{-5}
Training	
Batch size	40
Sequence length	20
Replay buffer capacity	250,000
Minimum buffer size	500
Imagination horizon (H)	15
Model training epochs	60
PPO epochs	5
Gradient clipping	100.0
Reinforcement Learning	
Discount factor (γ)	0.99
TD(λ) parameter	0.95
Entropy regularization	0.001
Initial Lagrangian multiplier (β_0)	1×10^{-4}

E Complete Baseline Comparison Results

This appendix provides complete performance comparisons between Safe Dreamers and MACPO-Discrete across all evaluated SMAC scenarios and cost limit configurations. The main paper presents results visually in Figure 5. This section includes all experimental configurations for completeness and reproducibility.

Table 3 presents the exhaustive evaluation results. For each scenario and cost limit, we report:

- **Score:** The mean cumulative reward.
- **Cost:** The mean number of allied deaths (casualty cost).
- **Win Rate:** The percentage of episodes won.

We compare Safe Dreamers (trained for 100k steps) against MACPO (trained for both 100k and 5M steps). This dual comparison highlights two key findings: first, matched at 100k steps, Safe Dreamers dominates MACPO in both safety and success; second, even when MACPO is given 50× more samples (5M steps), Safe Dreamers still achieves superior or comparable safety-performance trade-offs, particularly in harder scenarios like *8m* and *1c3s5z*.

Table 3: Complete performance comparison at 100k steps vs SafePO at 5M steps.

Algorithm	**Scenario**	**Score ↑**	**Cost ↓**	**Winrate ↑**	**Steps**	**Cost Limit**
MACPO	1c3s5z	8.72 ± 0.68	9.00 ± 0.00	0.00 ± 0.00	5M	0
MACPO	1c3s5z	7.64 ± 0.17	9.00 ± 0.00	0.00 ± 0.00	100k	0
Safe Dreamers	1c3s5z	**19.48 ± 0.35**	**3.40 ± 1.08**	**0.86 ± 0.10**	100k	0
MACPO	1c3s5z	10.07 ± 1.76	9.00 ± 0.00	0.00 ± 0.00	5M	4
MACPO	1c3s5z	7.79 ± 0.05	9.00 ± 0.00	0.00 ± 0.00	100k	4
Safe Dreamers	1c3s5z	**19.17 ± 0.21**	**4.94 ± 0.49**	**0.79 ± 0.03**	100k	4
MACPO	2m_vs_1z	11.74 ± 8.95	1.01 ± 1.08	0.49 ± 0.54	5M	0
MACPO	2m_vs_1z	2.71 ± 0.19	2.00 ± 0.00	0.00 ± 0.00	100k	0
Safe Dreamers	2m_vs_1z	**18.67 ± 0.29**	**0.27 ± 0.04**	**0.90 ± 0.02**	100k	0
MACPO	2m_vs_1z	12.05 ± 8.71	1.00 ± 1.09	0.50 ± 0.55	5M	1
MACPO	2m_vs_1z	3.13 ± 0.01	2.00 ± 0.00	0.00 ± 0.00	100k	1
Safe Dreamers	2m_vs_1z	**19.75 ± 0.19**	**0.14 ± 0.01**	**0.98 ± 0.02**	100k	1

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Table 3 – continued from previous page

Algorithm	Scenario	Score ↑	Cost ↓	Winrate ↑	Steps	Cost Limit
MACPO	2s3z	1.86 ± 1.96	5.00 ± 0.00	0.00 ± 0.00	5M	0
MACPO	2s3z	3.69 ± 0.02	5.00 ± 0.00	0.00 ± 0.00	100k	0
Safe Dreamers	2s3z	**18.29 ± 0.53**	**2.46 ± 0.29**	**0.71 ± 0.09**	100k	0
MACPO	2s3z	12.81 ± 7.41	3.47 ± 1.69	0.46 ± 0.51	5M	4
MACPO	2s3z	4.86 ± 0.04	5.00 ± 0.00	0.00 ± 0.00	100k	4
Safe Dreamers	2s3z	**18.11 ± 0.78**	**2.89 ± 0.17**	**0.69 ± 0.11**	100k	4
MACPO	2s_vs_1sc	2.51 ± 1.91	2.00 ± 0.00	0.00 ± 0.00	5M	0
MACPO	2s_vs_1sc	2.45 ± 0.12	2.00 ± 0.00	0.00 ± 0.00	100k	0
Safe Dreamers	2s_vs_1sc	**19.81 ± 0.13**	**0.25 ± 0.09**	**0.95 ± 0.02**	100k	0
MACPO	2s_vs_1sc	13.50 ± 7.32	1.08 ± 1.01	0.50 ± 0.54	5M	1
MACPO	2s_vs_1sc	4.21 ± 0.24	2.00 ± 0.00	0.00 ± 0.00	100k	1
Safe Dreamers	2s_vs_1sc	**19.97 ± 0.12**	**0.32 ± 0.15**	**0.97 ± 0.01**	100k	1
MACPO	3m	1.35 ± 1.43	3.00 ± 0.00	0.00 ± 0.00	5M	0
MACPO	3m	2.06 ± 0.12	3.00 ± 0.00	0.00 ± 0.00	100k	0
Safe Dreamers	3m	**19.32 ± 0.53**	**0.78 ± 0.14**	**0.95 ± 0.04**	100k	0
MACPO	3m	11.81 ± 8.96	1.64 ± 1.50	0.50 ± 0.55	5M	1
MACPO	3m	2.55 ± 0.04	3.00 ± 0.00	0.00 ± 0.00	100k	1
Safe Dreamers	3m	**19.12 ± 0.42**	**0.81 ± 0.15**	**0.93 ± 0.03**	100k	1
MACPO	3s5z_vs_3s6z	2.44 ± 1.10	8.00 ± 0.00	0.00 ± 0.00	5M	0
MACPO	3s5z_vs_3s6z	3.47 ± 0.03	8.00 ± 0.00	0.00 ± 0.00	100k	0
Safe Dreamers	3s5z_vs_3s6z	**12.89 ± 0.67**	**7.81 ± 0.25**	**0.06 ± 0.08**	100k	0
MACPO	3s5z_vs_3s6z	5.23 ± 0.27	8.00 ± 0.00	0.00 ± 0.00	5M	4
MACPO	3s5z_vs_3s6z	3.86 ± 0.04	8.00 ± 0.00	0.00 ± 0.00	100k	4
Safe Dreamers	3s5z_vs_3s6z	**11.31 ± 0.16**	**7.99 ± 0.02**	**0.00 ± 0.00**	100k	4
MACPO	3s_vs_3z	4.05 ± 3.23	3.00 ± 0.00	0.00 ± 0.00	5M	0
MACPO	3s_vs_3z	6.04 ± 0.05	3.00 ± 0.00	0.00 ± 0.00	100k	0
Safe Dreamers	3s_vs_3z	**19.83 ± 0.29**	**0.89 ± 0.22**	**0.95 ± 0.03**	100k	0
MACPO	3s_vs_3z	13.07 ± 6.56	1.91 ± 1.31	0.41 ± 0.47	5M	1
MACPO	3s_vs_3z	6.28 ± 0.11	3.00 ± 0.00	0.00 ± 0.00	100k	1
Safe Dreamers	3s_vs_3z	**19.56 ± 0.15**	**1.16 ± 0.32**	**0.92 ± 0.01**	100k	1
MACPO	3s_vs_4z	3.76 ± 1.29	3.00 ± 0.00	0.00 ± 0.00	5M	0
MACPO	3s_vs_4z	4.39 ± 0.04	3.00 ± 0.00	0.00 ± 0.00	100k	0
Safe Dreamers	3s_vs_4z	**20.39 ± 0.64**	**1.68 ± 0.39**	**0.83 ± 0.11**	100k	0
MACPO	3s_vs_4z	6.09 ± 1.45	3.00 ± 0.00	0.00 ± 0.00	5M	1
MACPO	3s_vs_4z	4.55 ± 0.01	3.00 ± 0.00	0.00 ± 0.00	100k	1
Safe Dreamers	3s_vs_4z	**19.68 ± 1.29**	**2.01 ± 0.38**	**0.67 ± 0.20**	100k	1
MACPO	3s_vs_5z	2.89 ± 0.93	3.00 ± 0.00	0.00 ± 0.00	5M	0
MACPO	3s_vs_5z	3.44 ± 0.03	3.00 ± 0.00	0.00 ± 0.00	100k	0
Safe Dreamers	3s_vs_5z	**19.81 ± 1.25**	**2.34 ± 0.51**	**0.44 ± 0.28**	100k	0
MACPO	3s_vs_5z	5.21 ± 1.56	3.00 ± 0.00	0.00 ± 0.00	5M	1
MACPO	3s_vs_5z	3.52 ± 0.02	3.00 ± 0.00	0.00 ± 0.00	100k	1
Safe Dreamers	3s_vs_5z	**17.42 ± 1.39**	**2.74 ± 0.14**	**0.19 ± 0.10**	100k	1
MACPO	8m	0.75 ± 0.62	8.00 ± 0.00	0.00 ± 0.00	5M	0
MACPO	8m	1.43 ± 0.01	8.00 ± 0.00	0.00 ± 0.00	100k	0
Safe Dreamers	8m	**19.01 ± 0.79**	**3.25 ± 1.14**	**0.88 ± 0.10**	100k	0
MACPO	8m	1.37 ± 0.58	8.00 ± 0.00	0.00 ± 0.00	5M	4
MACPO	8m	1.63 ± 0.01	8.00 ± 0.00	0.00 ± 0.00	100k	4

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Table 3 – continued from previous page

Algorithm	Scenario	Score ↑	Cost ↓	Winrate ↑	Steps	Cost Limit
Safe Dreamers	8m	**16.75 ± 1.29**	**5.25 ± 0.95**	**0.64 ± 0.13**	100k	4
MACPO	bane_vs_bane	**19.72 ± 0.31**	14.97 ± 7.87	**0.90 ± 0.10**	5M	0
MACPO	bane_vs_bane	19.26 ± 0.01	22.55 ± 0.02	0.79 ± 0.02	100k	0
Safe Dreamers	bane_vs_bane	19.13 ± 1.42	**11.33 ± 9.60**	0.80 ± 0.32	100k	0
MACPO	bane_vs_bane	19.64 ± 0.26	16.96 ± 6.11	0.89 ± 0.08	5M	4
MACPO	bane_vs_bane	19.37 ± 0.10	22.40 ± 0.07	0.80 ± 0.02	100k	4
Safe Dreamers	bane_vs_bane	**19.75 ± 0.41**	**10.88 ± 6.59**	**0.94 ± 0.09**	100k	4
MACPO	MMM	1.11 ± 0.72	10.00 ± 0.00	0.00 ± 0.00	5M	0
MACPO	MMM	1.87 ± 0.01	10.00 ± 0.00	0.00 ± 0.00	100k	0
Safe Dreamers	MMM	**17.24 ± 0.89**	**7.32 ± 0.64**	**0.58 ± 0.06**	100k	0
MACPO	MMM	1.16 ± 0.67	10.00 ± 0.00	0.00 ± 0.00	5M	4
MACPO	MMM	1.89 ± 0.04	10.00 ± 0.00	0.00 ± 0.00	100k	4
Safe Dreamers	MMM	**15.16 ± 1.24**	**7.96 ± 0.93**	**0.41 ± 0.13**	100k	4

F Cost Prediction Loss Function Ablation

During development, we observed that the world model's cost prediction was not sufficiently accurate. This highlights a key advantage of our approach: the world model enables direct inspection and debugging of cost predictions, allowing us to identify and address prediction errors. To improve cost prediction accuracy, we evaluated alternative loss functions beyond the standard mean squared error (MSE), including Huber loss, Focal loss, and Exponential loss. While these variants showed improvements in cost prediction accuracy, we retained the standard MSE loss in our final implementation for simplicity and robustness, maintaining focus on the main contribution. This ablation demonstrates that the framework is extensible and that the world model provides valuable diagnostic capabilities for iterative improvement—a key methodological advantage that enables principled debugging and optimization of individual components.

Table 4: Ablation: Cost prediction loss functions on 8M scenario. Results averaged over final training window (190k–240k steps). The world model's transparency enables direct inspection of prediction quality and loss, demonstrating that component-level debugging is more tractable than traditional black-box approaches.

Loss Function	Score ↑	Cost ↓	Win Rate (%) ↑	Model Loss ↓	Notes
Standard (MSE)	19.75	1.75	96.9	0.0094	Baseline, robust
Huber	19.89	1.31	98.7	0.0074	Best cost pred.
Focal	19.80	1.53	98.0	0.0084	Hard examples
Exponential	19.62	2.49	95.3	0.0102	Outlier sensitive

Huber reduces realized cost (1.31 vs. 1.75 for MSE) and model cost loss (0.0074), with a small score gain (19.89 vs. 19.75). Focal is competitive; Exponential underperforms (cost 2.49, loss 0.0102), likely due to outliers. Crucially, the world model's transparency lets us inspect `Model/cost_loss` and tie higher model loss to higher realized cost, enabling targeted component-level debugging. MSE remains a reasonable default; the framework is extensible and debuggable.

G Impact of Safety Mechanism

To verify the effectiveness of our safety components (Cost-Aware World Model and Lagrangian Update), we perform an ablation study comparing Safe Dreamers against an unconstrained baseline (equivalent to the original MAMBA algorithm [Egorov and Shpilman, 2022]). Figure 9 demonstrates the necessity of proactive safety: the unconstrained agent (red) consistently violates the cost limit, whereas Safe Dreamers (green) significantly reduces costs throughout training.

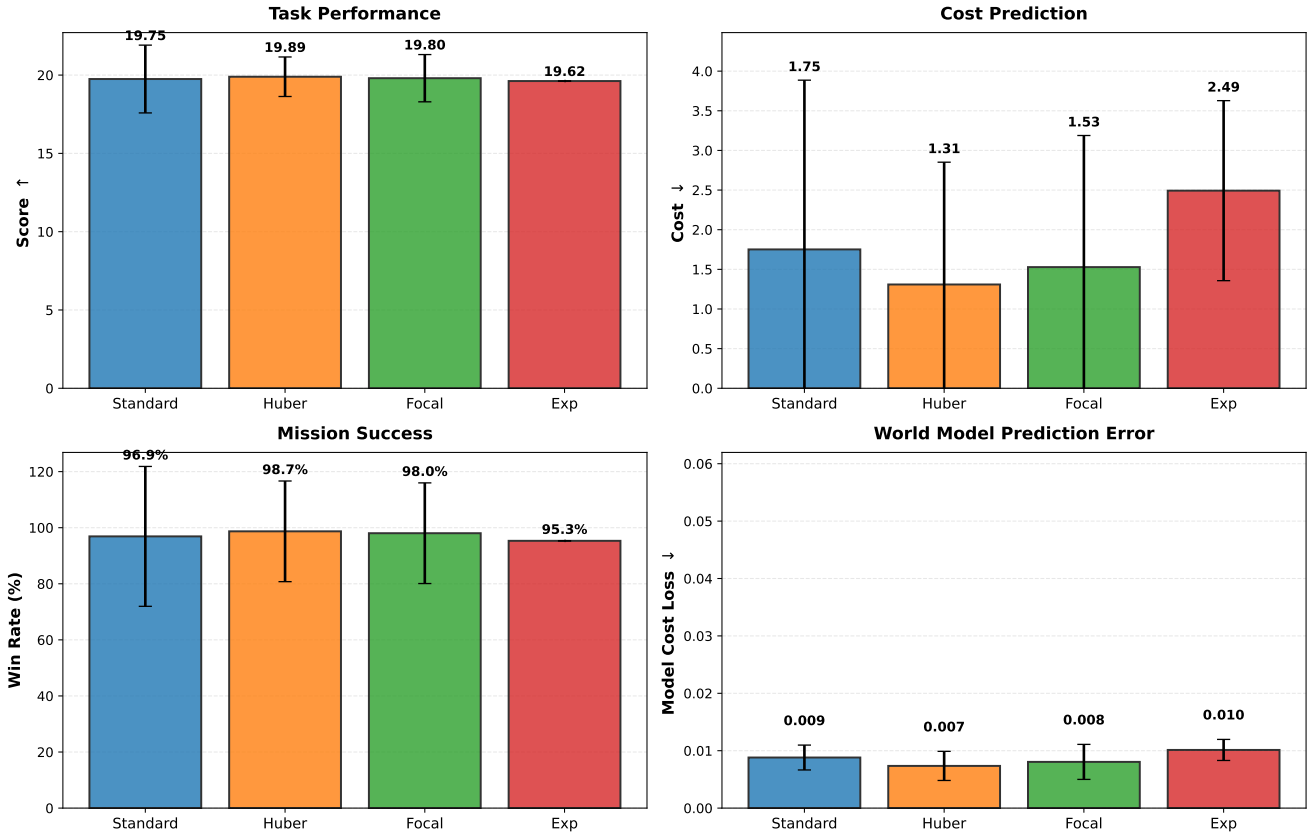
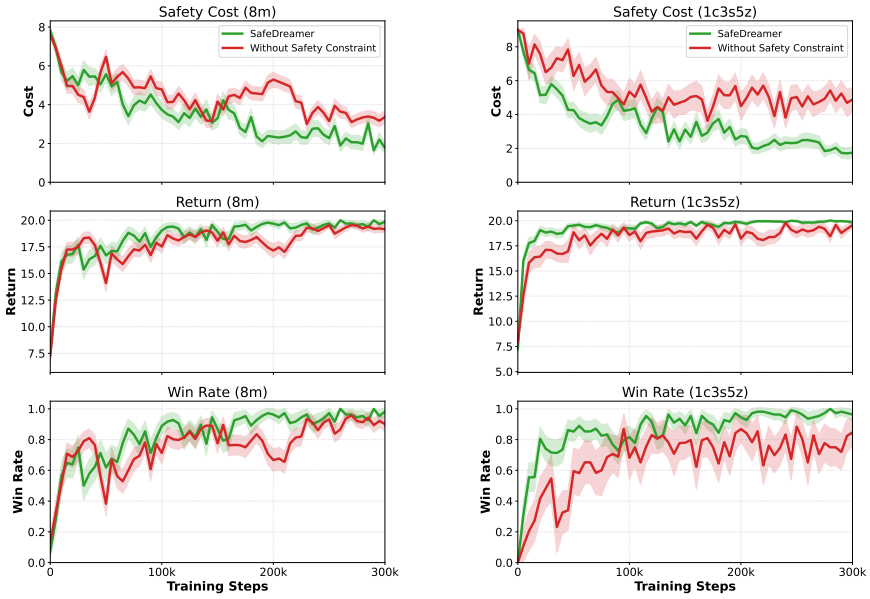


Figure 8: Cost prediction loss ablation across MSE, Huber, Focal, and Exponential. Huber consistently reduces cost and model loss with near-perfect success; Exponential underperforms. Differences are modest overall, so MSE remains a reasonable default. The world model makes these components inspectable and debuggable.



(a) 8m Scenario

(b) 1c3s5z Scenario

Figure 9: Ablation study demonstrating the effectiveness of the safety mechanism. We remove the safety layer and compare the unconstrained MAMBA agent against Safe Dreamers (cost limit $d = 0$) on the *8m* and *1c3s5z* scenarios. The green line represents Safe Dreamers, which significantly reduces the cost, while the red line shows higher costs for the unconstrained agent. Shaded regions represent the 95% confidence interval.